Linear regression lecture

We will start with a quick example, dataset that can be used instead of the Advertising data from Chapter 2, figure 2.1.

In particular this dataset record the:

- Sales
- Three variables, one response
- Similar statistics as the one for table 3.1 – 3.6

Motivating questions:

1. Is there an evidence of relationship between predictors and response?
2. How strong is the relationship between predictors and response? } How are strong is/based?
3. Which predictors are actually associated with the response? } Is 3 about all predictors quantitatively and 4 about their separately?
4. How strong is the relationship between each predictor and the response?
5. Is the relationship linear?
6. How accurately can we predict the response for new predictor values?

② Why can looker with the method.

Linear regression is an entry method to answering all these questions.

Notice most questions are about inference.

3.1 Simple linear regression

- Assume there's approximately a linear relationship between X and Y.
- Just for a single response Y and a single predictor X.

$Y \approx \beta_0 + \beta_1 X$ } this is our assumption for f the underlying ideal function that models the data

β_0 = intercept, β_1 = slope } coefficients or parameters = unknown constant that we want to estimate!

Once we have β_0, β_1, we can predict _____.

Check-in

- Interpreting what β_1, β_2 and all this notation mean using the example
- having people use the model to make a prediction } Similar to the paragraph below fig 3.1

What we will see:

1. How do we estimate the coefficients?
   - visual interpretation of residual sum of squares (RSS)
   - formula
   - goal is to minimize that, we can do it using calculus and we get 3.4

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})}$$

$$\hat{\beta}_0 = \bar{y} - \beta_1 \bar{x}$$

Why are we interested in these formulas?

- Answers best if they depend on the training data (x_i, y_i), different training data will give different estimates
- Your computer does these calculations

2. How accurate are the coefficients we found?

Accurate relative to what? To an assumption that the data is ideally described by $Y = \beta_0 + \beta_1 X$ (β_i are unknown)

Standard errors give us a way to quantify the average amount that our estimates $\hat{\beta}_i$ differ from the actual values β_i.

*standard error formulas and assumptions

What to get out of these formulas?

- There's one standard error per coefficient, not one per model.
- σ^2 is not known so it's estimated from the data as well

Confidence intervals can be computed using SEs.

95% confidence interval = range of values s.t. with 95% probability, the range will contain the true unknown value of the parameter. * go back to Discrete notes.

Check-in

- A small exercise about confidence interval and how to interpret it.
- Two extreme cases of a really big one and a very small one, relative to X units.
- Remember you can never estimate a parameter as a single number, there's always uncertainty associated with it.

For the simple linear regression the confidence interval is given by

$$[\hat{\beta}_i - 2 \cdot SE(\hat{\beta}_i), \hat{\beta}_i + 2 \cdot SE(\hat{\beta}_i)]$$

Interpreting confidence interval for β_0

95% confidence interval for β_0 is $[a_0, b_0]$ means that there's approx a 95% chance that the linear model will intercept the x-axis between a_0 and b_0

In our example: in the absence of any _X_, _Y_ will, on average, fall somewhere between a_0 and b_0 units.

Interpreting confidence interval for β_1

95% confidence interval for β_1 is $[a_1, b_1]$ means that there's approx a 95% chance that the slope will be between a_1 and b_1.

In our example: for an average increase of _X_, there will be an average increase in _Y_ between a_1 and b_1 units.

★ What to get from this discussion?

Rank these from best to worst about how to communicate your results:

- For every X units we will gain y.
- For every X units we estimate we will gain y.
- For every X units we estimate with high probability that we will gain between a_1 and b_1 of y.

Which would you prefer?

Is there a relationship between X and Y?

H_0 = null hypothesis = There is no relationship between X and Y. : $\beta_1 = 0$

H_a = alternative hypothesis = There is some relationship between X and Y : $\beta_1 \neq 0$

If $\beta_1 = 0$: $Y = \beta_0$: there's no X affecting the Y!

If $\beta_1 \neq 0$: $Y = \beta_0 + \beta_1 X$: Y is related to X

To test if $\beta_1 = 0$, we need to determine if $\hat{\beta}_1$ (estimate for β_1) is sufficiently far from 0.

How this actually works (though we unfortunately not cover it in class)

1. compute t-statistic $t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$ (3.14)

   = number of standard deviations that $\hat{\beta}_1$ is away from 0

2. Assuming $\beta_1 = 0$ (no relationship between X and Y) and $n > 30$, then t's distribution is very similar to normal distribution
3. So we can compute the probability of observing any number that is an absolute value greater than $|t|$.

   ↳ "What is the probability that $|\hat{\beta}_1|$ is more than t standard deviations away from zero if the true parameter is $\beta_1 = 0$?"

Option 1: small probability that $|\hat{\beta}_1|$ is that far away from 0 if there's no relation ⇒ it must be that $\beta_1 \neq 0$

Option 2: high probability that $|\hat{\beta}_1|$ is that far away from 0 if there's no relation ⇒ maybe $\beta_1 = 0$, we cannot discard this possibility.

We can use p-values to test this.

- small p-values ⇒ small probability that $|\hat{\beta}_1|$ is that far away from 0
  - ⇒ it must be that $\beta_1 \neq 0$
  - ⇒ it must be that X is related to Y

Check-in

	Coefficient	Std error	p-value
intercept	$\hat{\beta}_0$ = 7.0325	0.4578	< 0.0001
Y	$\hat{\beta}_1$ = 0.0475	0.0027	< 0.0001
our estimates	used to compute conf. intervals & p-values		probability of seeing these estimates if $\beta_i = 0$

The probabilities of seeing these values for the coefficients β_0 and β_1 if there is no relationship between X and Y are virtually zero.

- small p-value for β_1 ⇒ there is a relationship between X and Y
- small p-value for β_0 ⇒ if $X = 0$, then $Y \neq 0$ } this might need to be separately explained as another thing.

How accurate is our linear regression model?

We will cover two ways of quantifying it:

- Residual standard error
- R^2

Residual standard error (RSE)

average amount that the response $\hat{Y}$ will deviate from the true regression line.

$$RSE = \sqrt{\frac{1}{n-2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \quad (3.15)$$

What would indicate a better fit, a small or large RSE?

A small RSE would come from $y_i - \hat{y}_i$ being very close to 0

⇒ (actual value) − (estimate) very close to 0

⇒ actual value ≈ estimate "very close" or "approximately"

Challenges: measured in the units of Y, it can be hard to know what's a "small" RSE.

R^2 statistic

- A proportion: always between 0 and 1 → easier to interpret.

total sum of squares: $TSS = \sum (y_i - \bar{y})^2$: variability in Y

residual sum of squares: $RSS = \sum (y_i - \hat{y}_i)^2$: variability in Y unexplained by the regression "explained by regression" means $\hat{y}_i = \bar{y}_i$.

So...

$$\frac{(\text{variability in } Y) - (\text{variability in } Y \text{ unexplained by regression})}{\text{variability in } Y} = \text{proportion of variability in } Y \text{ explained by regression}$$

$$R^2 = \frac{TSS - RSS}{TSS} = \text{proportion of variability in } Y \text{ that can be explained using } X$$

R^2 close to 0 ⇒ almost none of the variability in Y can be explained by X

- why? – linear model is not a good option
  - error variance is huge
  - both

Interpretation still depends on the application.

A note about correlation: $Cor(X, Y)$ also measures the relationship between X and Y in two variables $R^2 = Cor(X, Y)^2$, but R^2 is easier to generalize to more than two variables.